



AX7203 Development Board User Manual

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Document revision history

chapter	Revision Overview
V1.0	
all	First version released

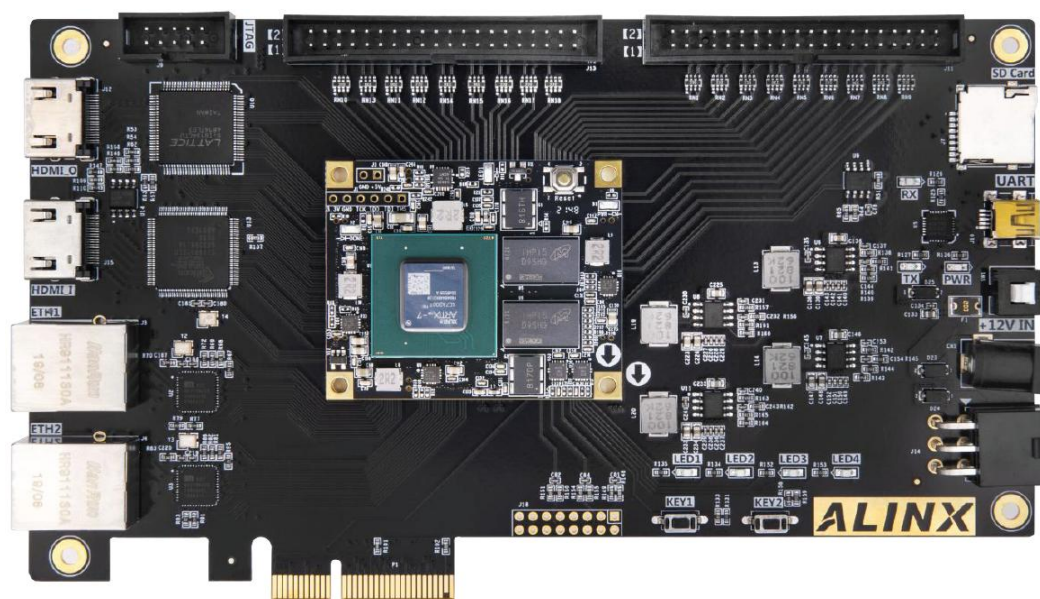
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The ARTIX-7 FPGA development platform (model: AX7203) has been officially released . This user manual has been written to help you quickly understand this development platform .

This ARTIX-7 FPGA development platform employs a SoM and expansion board architecture, facilitating users' secondary development of the SoM. The carrier board design features a rich array of peripheral interfaces, including one PCIe x4 interface , two Gigabit Ethernet interfaces , one HDMI output interface , one HDMI input interface , a UART interface , and an SD card interface . These interfaces meet various requirements for high-speed PCIe data transmission , video image processing, and industrial control. It provides possibilities for early verification and later applications of high-speed video transmission, network and PCIe communication, and data processing.



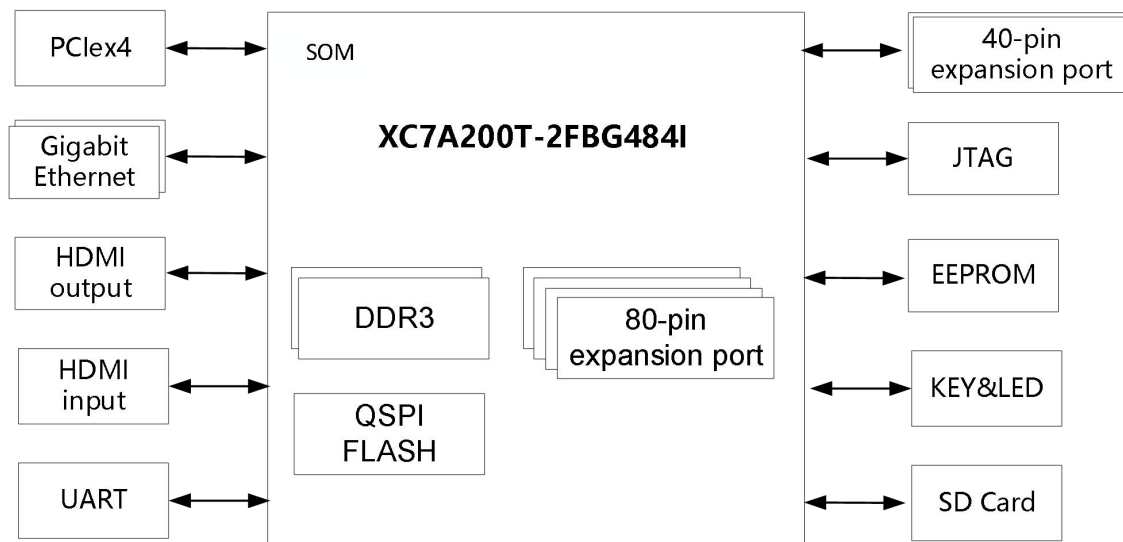
Chapter 1 Development Board Introduction

The development board is designed with a SoM and a carrier board . The SoM and the carrier board are connected using a high-speed board-to-board connector .

The SoM mainly consists of an FPGA, two DDR3 memory chips, and a QSPI FLASH, handling high-speed data processing and storage for the FPGA . Combined with high-speed data read/write between the FPGA and the two DDR3 chips, the data bit width is 32 bit, resulting in a total system bandwidth of 25Gb/s (800M*32bit). Furthermore , the two DDR3 chips have a capacity of 8Gbit, meeting the high buffer requirements during data processing. The FPGA we selected is the Xilinx ARTIX-7 series XC7A200T chip , packaged in an FBG484 package. The communication clock frequency between the XC7A200T and the DDR3 chips reaches 400MHz , with a data rate of 800MHz , fully meeting the requirements for high-speed multi-channel data processing. Additionally, the XC7A200T FPGA features four GTP high-speed transceivers, each with a speed of up to 6.6Gb/s, suitable for fiber optic communication and PCIe data communication.

The carrier board provides the SoM with a rich set of peripheral interfaces, including a PCIe x4 interface, two Gigabit Ethernet interfaces, one HDMI output interface, one HDMI input interface, one UART serial port interface, one SD card interface, one XADC connector interface, two 40-pin expansion ports , as well as some buttons , LEDs and EEPROM circuits.

The following diagram shows the overall structure of the development system:



This diagram illustrates the functionality that our development platform can achieve.

- The Artix-7 FPGA SoM consists of an XC7A200T processor , 8Gb DDR3 memory, and 128Mb QSPI FLASH memory. It also includes two high-precision Sitime LVDS differential crystal oscillators, one at 200MHz and the other at 125MHz, providing stable clock input for the FPGA system and GTP module.
- 1 PCIe x4 interface : Supports PCI Express 2.0 standard, provides PCIe x4 high-speed data

transmission interface, and single-channel communication rate can reach up to 5GBaud .

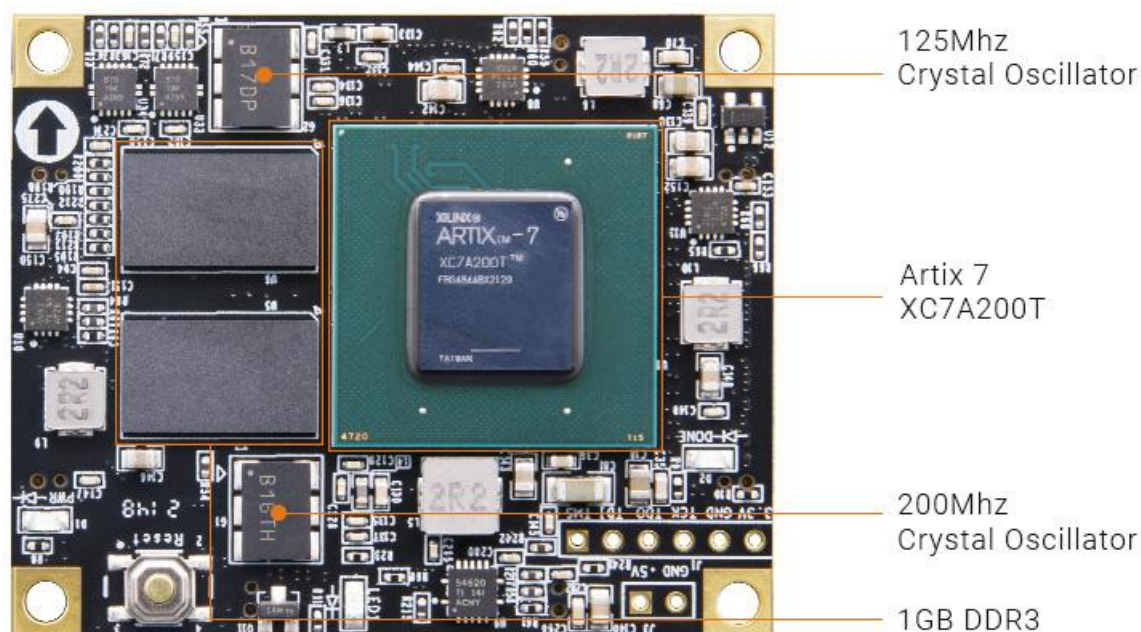
- Two 10 /100M /1000M Ethernet RJ-45 interfaces : The Gigabit Ethernet interface chip uses the KSZ9031 industrial-grade GPHY chip to provide network communication services for users. The KSZ9031 chip supports 10/100/1000 Mbps network transmission rates , full-duplex, and auto-negotiation.
- One HDMI output : We selected the SIL9134 HDMI encoding chip from Silicon Image , which supports up to 1080P@60Hz output and 3D output.
- One HDMI input : We selected the Silicon Image SIL9013 HDMI decoding chip, which supports up to 1080P@60Hz input and supports data output in different formats.
- One USB UART interface : One UART to USB interface for communication with a computer, facilitating user debugging. The serial port chip uses the Silicon Labs CP2102GM USB-UART chip , and the USB interface uses a MINI USB interface.
- Micro SD Card Slot : One Micro SD card slot, supporting SD mode and SPI mode.
- EEPROM : One IIC interface EEPROM 24LC04 .
- Two 40-pin expansion ports : Two 40-pin, 2.54mm pitch expansion ports are provided for connecting various Alinx modules (dual-lens cameras, TFT LCD screens, high-speed AD modules, etc.). The expansion ports include one 5V power supply, two 3.3V power supply ports, three ground ports, and 34 I/O ports.
- JTAG port : A 10-pin 2.54mm standard JTAG port used for downloading and debugging FPGA programs .
- Buttons : 2 user buttons, 1 reset button (on the SoM) .
- LED lights : 5 user-defined light -emitting diodes (1 on the SoM and 4 on the carrier board) ;

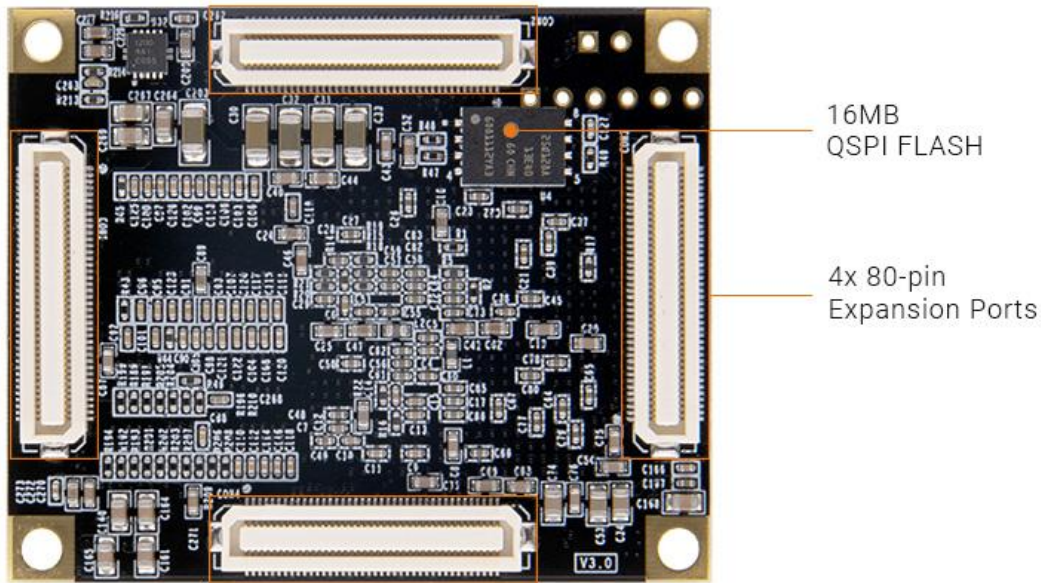
Chapter 2 FPGA

2.1 Introduction

The AC7200 SoM is a high-performance SoM developed based on the XILINX ARTIX-7 series XC7A200T-2FBG484I chip. It features high speed, high bandwidth , and high capacity, making it suitable for high-speed data communication, video image processing, and high-speed data acquisition.

This SoM uses two Micron MT41J256M16HA-125 chips . The DDR3 chips, each with a capacity of 512MB , are combined in pairs to form a 32-bit data bus, achieving a read/write bandwidth of up to 25Gb/s between the FPGA and DDR3 . This configuration can meet the demands of high-bandwidth data processing . The SoM expands to include 180 standard 3.3V I/O ports, 15 standard 1.5V I/O ports, and 4 pairs of GTP high-speed RX/TX differential signals. This SoM is a good choice for users requiring a large number of I/O pins. The traces between the FPGA chips and the interfaces are designed with equal length and differential routing, and the SoM's dimensions of 45*55 (mm) make it ideal for secondary development.





2.2 FPGA

The FPGA we used is model XC7A200T-2FBG484I, belonging to Xilinx's Artix-7 series, with a speed class of 2 and an industrial-grade temperature class. This model is in an FBG484 package with 484 pins. The naming convention for Xilinx ARTIX-7 FPGA chips is as follows:

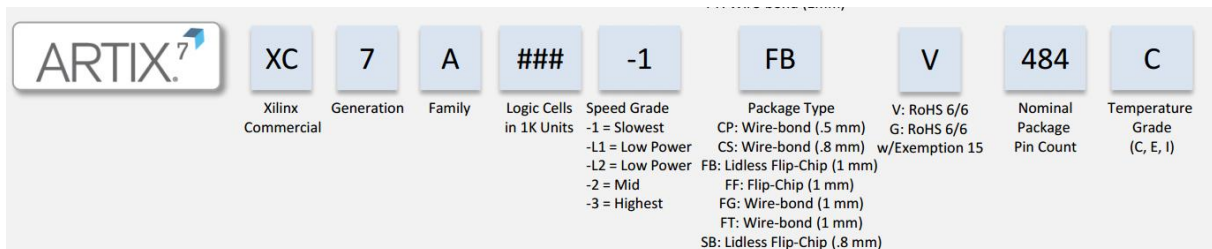


Figure 2.2.1 Chip Naming Conventions

The main parameters of the FPGA chip XC7A200T are as follows:

Name	Specific parameters
Logic Cells	215360
Lookup Table (Slices)	33650
Trigger (CLB flip-flops)	269200
Block RAM (kb) size	13140
DSP processing unit (DSP Slices)	740
PCIe Gen2	Supports PCI Express 2.0 x4
Analog-to-digital conversion/XADC	One 12-bit, 1Mbps AD converter

Name	Specific parameters
GTP Transceiver	4, 6.6Gb/s max
Speed level	-2
Temperature rating	Industrial grade

Table 2.2.1 - Main Parameters of the Chip

FPGA power supply system

The Artix-7 FPGA power supply includes VCCINT, VCCBRAM, VCCAUX, VCCO, VMGTAVCC, and VMGTAVTT. VCCINT is the power supply pin for the FPGA core and should be connected to 1.0 V; VCCBRAM is the power supply pin for the FPGA Block RAM and should be connected to 1.0 V; VCCAUX is the auxiliary power supply pin for the FPGA and should be connected to 1.8 V; VCCO provides the supply voltage for all FPGA BANKs – including BANK0, BANK13–16, and BANK34–35. On the AC7200 core board, BANK34 and BANK35 require connection to DDR3 memory; therefore, their supply voltage is set to 1.5 V, while the other BANKs are powered at 3.3 V. The VCCO voltage for BANK15 and BANK16 is supplied by an LDO; this voltage level can be adjusted by replacing the LDO chip. VMGTAVCC is the power supply voltage for the FPGA's internal GTP transceiver and should be connected to 1.0 V; VMGTAVTT is the termination voltage for the GTP transceiver and should be connected to 1.2 V.

For the Artix-7 FPGA system, the power-up sequence is: first VCCINT, then VCCBRAM, followed by VCCAUX, and finally VCCO. If the voltages of VCCINT and VCCBRAM are identical, they can be powered on simultaneously; the power-down sequence is the reverse. For the GTP transceiver, the power-up sequence is: VCCINT, then VMGTAVCC, and finally VMGTAVTT. If the voltages of VCCINT and VMGTAVCC are identical, they can be powered on simultaneously; the power-down sequence is exactly the reverse of the power-up sequence.

2.3 Active differential crystal oscillator

The AC7200 SoM is equipped with two active differential crystal oscillators from Sitime: one is a 200MHz SiT9102 , used as the system master clock for the FPGA and for generating the DDR3 control clock; the other is 125MHz SiT9102 , used as the reference clock input for the GTP transceiver.

1) 200MHz differential clock

In the schematic, G1 is a 200MHz active differential crystal oscillator circuit that provides the system clock source for the development board . The crystal oscillator output is connected to the BANK34 global clock pin MRCC (R4 and T4) of the FPGA . This 200MHz differential clock can be used to drive user logic circuits within the FPGA . Users can generate clocks of different frequencies by configuring the PLLs and DCMs inside the FPGA.

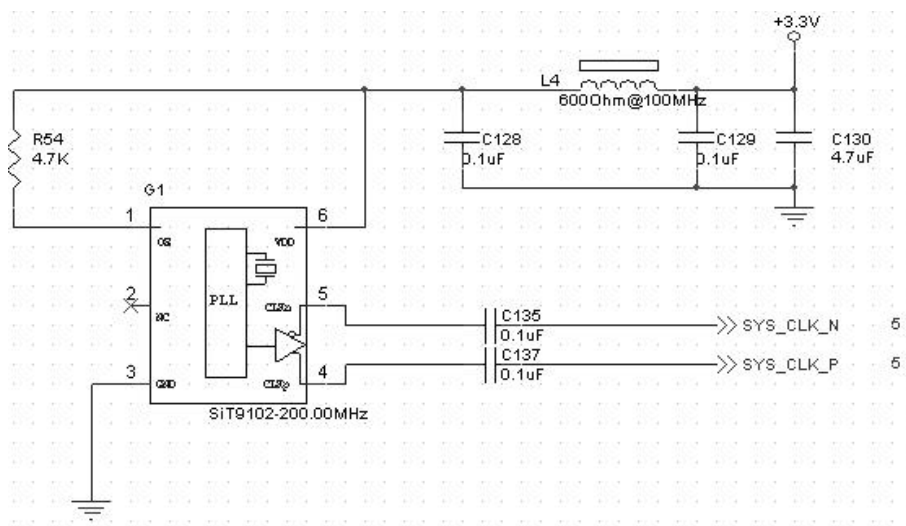


Figure 2.3.1 - 200MHz Active Differential Crystal Oscillator

Clock pin assignment:

Pin Name	FPGA pins
SYS_CLK_P	R4
SYS_CLK_N	T4

Table 2.3.1 - 200MHz Clock Pin Assignment

2) 125MHz differential clock

In Figure 2.3.2 , G2 is a 125MHz active differential crystal oscillator circuit. This clock provides the reference input clock for the GTP module inside the FPGA. The crystal oscillator output is connected to the BANK216 clock pins MGTREFCLK0P (F6) and MGTREFCLK0N (E6) of the FPGA GTP .

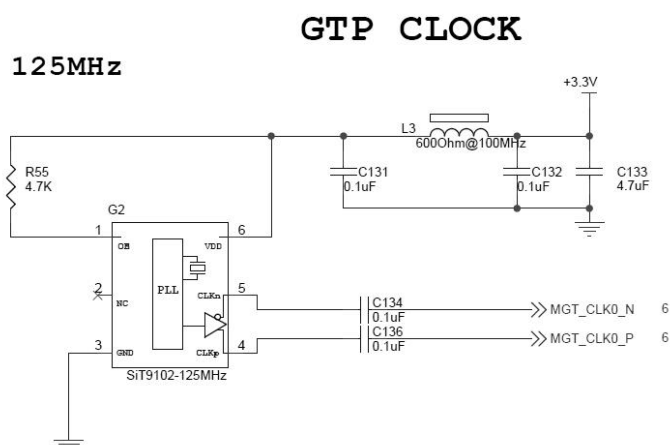


Figure 2.3.2-125MHz active differential crystal oscillator

Clock pin assignment:

Pin Name	FPGA pins
MGT_CLK0_P	F6
MGT_CLK0_N	E6

Table 2.3.2 - 125MHz Clock Pin Assignment

2.4 DDR3

The AC7200 SoM features two Micron 4Gbit (512 MB) DDR3 chips (totaling 8Gbit) , model MT41J256M16HA-125 (compatible with MT41K256M16HA-125) . The DDR bus width is 32 bits. The maximum operating clock speed of the DDR3 SDRAM is 800 MHz (data rate 1600 Mbps) . This DDR3 memory system is directly connected to the memory interfaces of FPGA 's BANK 34 and BANK 35. The specific configuration of the DDR3 SDRAM is shown in Table 2.4.1 below .

Position	Chip type	Capacity	Factory
U5 , U6	MT41J256M16HA-125	256 M x 16bit	Micron

Table 2.4.1 - DDR3 SDRAM Configuration

The hardware design of DDR3 requires strict consideration of signal integrity. During circuit and PCB design, we have fully considered matching resistors/terminating resistors , trace impedance control , and trace length control to ensure stable high-speed operation of DDR3 . A hardware connection diagram of DDR3 DRAM is shown in Figure 2.4.1 :

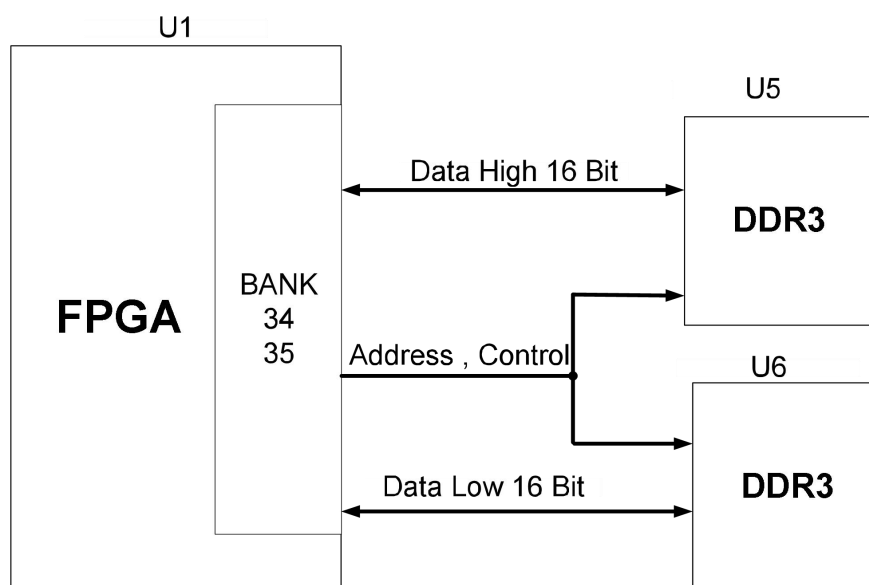


Figure 2.4.1 - DDR3 DRAM Schematic Diagram

DDR3 DRAM pin assignment:

Signal name	FPGA pin names	FPGA pin numbers
DDR3_DQS0_P	IO_L3P_T0_DQS_AD5P_35	E1

Signal name	FPGA pin names	FPGA pin numbers
DDR3_DQS0_N	IO_L3N_T0_DQS_AD5N_35	D1
DDR3_DQS1_P	IO_L9P_T1_DQS_AD7P_35	K2
DDR3_DQS1_N	IO_L9N_T1_DQS_AD7N_35	J2
DDR3_DQS2_P	IO_L15P_T2_DQS_35	M1
DDR3_DQS2_N	IO_L15N_T2_DQS_35	L1
DDR3_DQS3_P	IO_L21P_T3_DQS_35	P5
DDR3_DQS3_N	IO_L21N_T3_DQS_35	P4
DDR3_D Q [0]	IO_L2P_T0_AD12P_35	C2
DDR3_D Q [1]	IO_L5P_T0_AD13P_35	G1
DDR3_D Q [2]	IO_L1N_T0_AD4N_35	A1
DDR3_D Q [3]	IO_L6P_T0_35	F3
DDR3_D Q [4]	IO_L2N_T0_AD12N_35	B2
DDR3_D Q [5]	IO_L5N_T0_AD13N_35	F1
DDR3_D Q [6]	IO_L1P_T0_AD4P_35	B1
DDR3_D Q [7]	IO_L4P_T0_35	E2
DDR3_D Q [8]	IO_L11P_T1_SRCC_35	H3
DDR3_DQ [9]	IO_L11N_T1_SRCC_35	G3
DDR3_DQ [10]	IO_L8P_T1_AD14P_35	H2
DDR3_DQ [11]	IO_L10N_T1_AD15N_35	H5
DDR3_DQ [12]	IO_L7N_T1_AD6N_35	J1
DDR3_DQ [13]	IO_L10P_T1_AD15P_35	J5
DDR3_DQ [14]	IO_L7P_T1_AD6P_35	K1
DDR3_DQ [15]	IO_L12P_T1_MRCC_35	H4
DDR3_DQ [16]	IO_L18N_T2_35	L4
DDR3_DQ [17]	IO_L16P_T2_35	M3
DDR3_DQ [18]	IO_L14P_T2_SRCC_35	L3
DDR3_DQ [19]	IO_L17N_T2_35	J6
DDR3_DQ [20]	IO_L14N_T2_SRCC_35	K3
DDR3_DQ [21]	IO_L17P_T2_35	K6

Signal name	FPGA pin names	FPGA pin numbers
DDR3_DQ [22]	IO_L13N_T2_MRCC_35	J4
DDR3_D Q [23]	IO_L18P_T2_35	L5
DDR3_D Q [24]	IO_L20N_T3_35	P1
DDR3_D Q [25]	IO_L19P_T3_35	N4
DDR3_D Q [26]	IO_L20P_T3_35	R1
DDR3_D Q [27]	IO_L22N_T3_35	N2
DDR3_D Q [28]	IO_L23P_T3_35	M6
DDR3_D Q [29]	IO_L24N_T3_35	N5
DDR3_D Q [30]	IO_L24P_T3_35	P6
DDR3_D Q [31]	IO_L22P_T3_35	P2
DDR3_DM0	IO_L4N_T0_35	D2
DDR3_DM1	IO_L8N_T1_AD14N_35	G2
DDR3_DM2	IO_L16N_T2_35	M2
DDR3_DM3	IO_L23N_T3_35	M5
DDR3_A[0]	IO_L11N_T1_SRCC_34	AA4
DDR3_A[1]	IO_L8N_T1_34	AB2
DDR3_A[2]	IO_L10P_T1_34	AA5
DDR3_A[3]	IO_L10N_T1_34	AB5
DDR3_A[4]	IO_L7N_T1_34	AB1
DDR3_A[5]	IO_L6P_T0_34	U3
DDR3_A[6]	IO_L5P_T0_34	W1
DDR3_A[7]	IO_L1P_T0_34	T1
DDR3_A[8]	IO_L2N_T0_34	V2
DDR3_A[9]	IO_L2P_T0_34	U2
DDR3_A[10]	IO_L5N_T0_34	Y1
DDR3_A[11]	IO_L4P_T0_34	W2
DDR3_A[12]	IO_L4N_T0_34	Y2
DDR3_A[13]	IO_L1N_T0_34	U1
DDR3_A[14]	IO_L6N_T0_VREF_34	V3

Signal name	FPGA pin names	FPGA pin numbers
DDR3_BA[0]	IO_L9N_T1_DQS_34	AA3
DDR3 _ BA [1]	IO_L9P_T1_DQS_34	Y3
DDR3 _ BA [2]	IO_L11P_T1_SRCC_34	Y4
DDR3 _ S0	IO_L8P_T1_34	AB3
DDR3 _ RAS	IO_L12P_T1_MRCC_34	V4
DDR3 _ CAS	IO_L12N_T1_MRCC_34	W4
DDR3 _ WE	IO_L7P_T1_34	AA1
DDR3 _ ODT	IO_L14N_T2_SRCC_34	U5
DDR3 RESET	IO_L15P_T2_DQS_34	W6
DDR3_CLK_P	IO_L3P_T0_DQS_34	R3
DDR3_CLK_N	IO_L3N_T0_DQS_34	R2
DDR3_CKE	IO_L14P_T2_SRCC_34	T5

Table 2.4.2 - DDR3 DRAM Pin Assignment

2.5 QSPI Flash

The SoM uses a 128 Mbit QSPI FLASH chip, model N25Q128 , which uses the 3.3V CMOS voltage standard. Due to its non-volatile nature, the QSPI FLASH can be used as a boot image for the FPGA system. These images mainly include the FPGA 's bit file, the soft-core application code, and other user data files. The specific model and related parameters of the SPI FLASH are shown in the table below .

Position	Chip type	Capacity	Factory
U8	N25Q128	128Mbit	Numonyx

Table 2.5.1 - QSPI Flash Models and Parameters

BANK 0 and BANK14 of the FPGA chip . The clock pin is connected to CCLK0 of BANK0, and the other data and chip select signals are connected to D00~D03 and FCS pins of BANK14, respectively . Figure 2.5.1 is a schematic diagram of the QSPI Flash hardware connection.

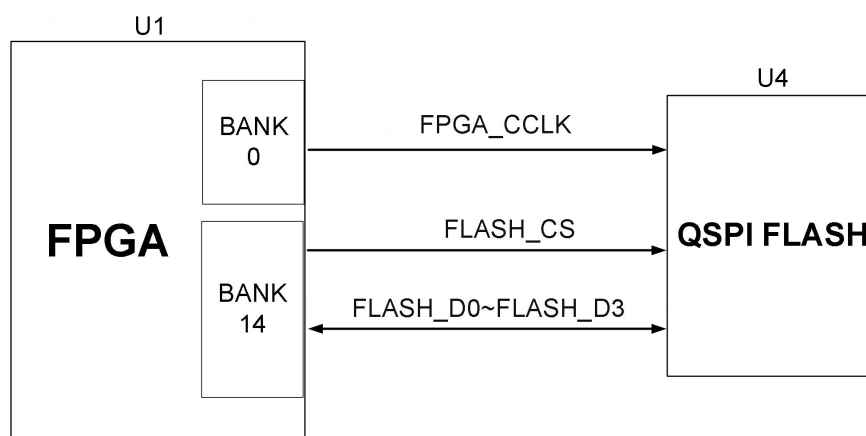


Figure 2.5.1 - QSPI Flash Connection Diagram

Configure chip pin assignments:

Signal name	FPGA pin names	FPGA pin numbers
QSPI_CLK	CCLK_0	L12
QSPI_CS	IO_L6P_T0_FCS_B_14	T19
QSPI_DQ0	IO_L1P_T0_D00_MOSI_14	P22
QSPI_DQ1	IO_L1N_T0_D01_DIN_14	R22
QSPI_DQ2	IO_L2P_T0_D02_14	P21
QSPI_DQ3	IO_L2N_T0_D03_14	R21

Table 2.5.2 - QSPI Flash Pin Assignment

2.6 LED

The AC7200 SoM has three red LEDs: one is the power indicator (PWR) , one is the configuration LED (DONE) , and the other is the user LED (LED1). The power indicator lights up when the SoM is powered on; the configuration LED lights up after the FPGA configuration program is executed. The user LED is connected to the BANK34's I/O pin and can be controlled programmatically to turn on and off. The user LED lights up when the I/O voltage connected to it is high and turns off when the I/O voltage is low . A schematic diagram of the LED hardware connection is shown in Figure 2.6.1 .

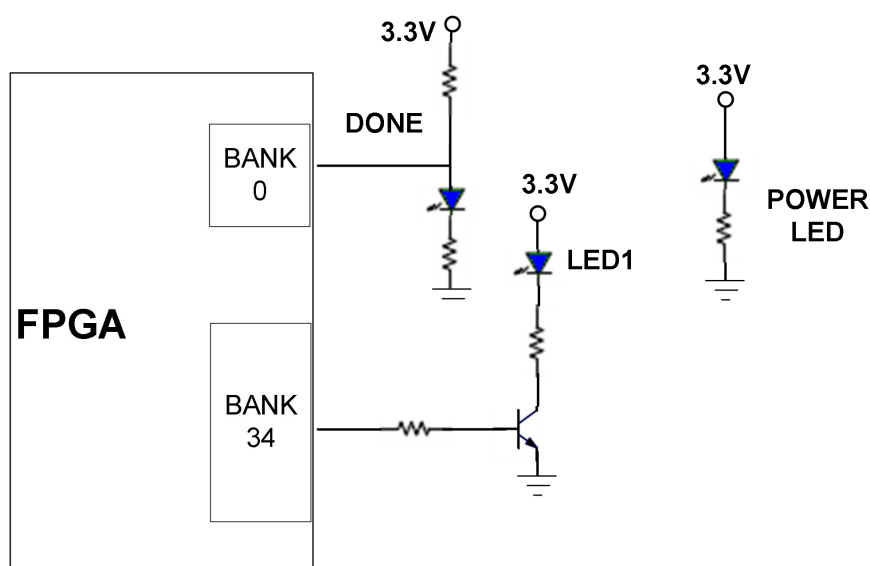


Figure 2.6.1 - Hardware connection diagram of LED light on development board

Pin assignment for user LED lights

Signal name	FPGA pin names	FPGA pin numbers	Remark
LED1	IO_L15N_T2_DQS_34	W5	User LED lights

Table 2.6.1 - Pin Assignments for User LEDs

2.7 Reset Button

The AC7200 development board has a reset button, which is connected to a general-purpose I/O pin on the FPGA chip's BANK34. Users can use this reset button to initialize the FPGA program. In the design, when the button is pressed, the signal voltage input to the I/O pin is low, and the reset signal is valid; when the button is not pressed, the signal input to the I/O pin is high. A schematic diagram of the reset button connection is shown in Figure 2.7.1 .

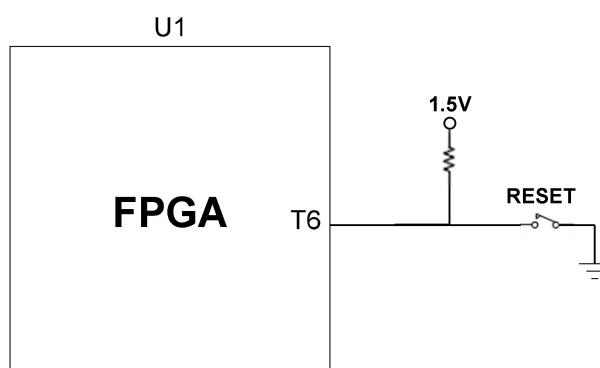


Figure 2.7.1 - Schematic diagram of reset button connection

Pin assignment for the reset button

Signal name	FPGA pin names	FPGA pin numbers	Remark
RESET_N	IO_L17N_T2_34	T6	Reset button

Table 2.7.1 - Pin Assignment for Reset Button

2.8 JTAG Interface

We have also reserved a JTAG test socket J1 on the AC7200 SoM for independent JTAG downloading and debugging on the SoM. The JTAG interface J1 on the SoM uses a single row of 6-pin test holes with a 2.54mm pitch. If users need to use JTAG connection for debugging on the SoM, they need to solder a single row of 6-pin headers.

2.9 Power Interface

In order to enable the SoM to work independently, we have reserved a 2-pin power interface (J3) for the SoM. When the user supplies power to the SoM through J3, the power supply cannot be supplied through the carrier board , otherwise it will cause current conflict.

2.10 Extension Interface

The back of the SoM has four high-speed expansion ports, which are connected to the carrier board using four 80-pin board - to -board connectors . The FPGA's I/O ports are connected to these four expansion ports via differential routing. The connector pin pitch is 0.5mm, and the configuration with the female connector on the carrier board enables high-speed data communication.

Expansion port CON1

The 80-pin connector CON1 is used to connect the VCCIN power supply (+5V) of the carrier board , ground, and the general-purpose I/O of the FPGA. Note that CON1 has 15 pins connected to the BANK34 I/O ports. Because the BANK34 connection is to DDR3, all BANK34 I/O ports are rated for 1.5V. The pin assignments for the CON1 expansion port are shown in Table 2.10.1 .

CON1 pin	Signal name	FPGA pin number	Level Standard	CON1 pin	Signal name	FPGA pin number	Level Standard
PIN1	VCCIN	-	+5V	PIN2	VCCIN	-	+5V
PIN3	VCCIN	-	+5V	PIN4	VCCIN	-	+5V
PIN5	VCCIN	-	+5V	PIN6	VCCIN	-	+5V
PIN7	VCCIN	-	+5V	PIN8	VCCIN	-	+5V
PIN9	GND	-	GND	PIN10	GND	-	GND
PIN11	NC	-	NC	PIN12	NC	-	NC
PIN13	NC	-	NC	PIN14	NC	-	NC
PIN15	NC	-	NC	PIN16	B13_L4_P	AA15	3.3V

CON1 pin	Signal name	FPGA pin number	Level Standard	CON1 pin	Signal name	FPGA pin number	Level Standard
PIN17	NC	-	NC	PIN18	B13_L4_N	A B 15	3.3V
PIN19	GND	-	GND	PIN20	GND	-	GND
PIN21	B13_L5_P	Y13	3.3V	PIN22	B13_L1_P	Y16	3.3V
PIN23	B13_L5_N	AA14	3.3V	PIN24	B13_L1_N	AA16	3.3V
PIN25	B13_L7_P	AB11	3.3V	PIN26	B13_L2_P	AB16	3.3V
PIN27	B13_L7_P	AB1 2	3.3V	PIN28	B13_L2_N	AB1 7	3.3V
PIN29	GND	-	GND	PIN30	GND	-	GND
PIN31	B13_L3_P	AA13	3.3V	PIN32	B13_L6_P	W14	3.3V
PIN33	B13_L3_N	A B 13	3.3V	PIN34	B13_L6_N	Y14	3.3V
PIN35	B34_L23_P	Y8	1.5V	PIN36	B34_L20_P	AB7	1.5V
PIN37	B34_L23_N	Y 7	1.5V	PIN38	B34_L20_N	AB 6	1.5V
PIN39	GND	-	GND	PIN40	GND	-	GND
PIN41	B34_L18_N	AA6	1.5V	PIN42	B34_L21_N	V8	1.5V
PIN43	B34_L18_P	Y6	1.5V	PIN44	B34_L21_P	V9	1.5V
PIN45	B34_L19_P	V7	1.5V	PIN46	B34_L22_P	AA8	1.5V
PIN47	B34_L19_N	W7	1.5V	PIN48	B34_L22_N	AB8	1.5V
PIN49	GND	-	GND	PIN50	GND	-	GND
PIN51	XADC_VN	M9	simulation	PIN52	NC		
PIN53	XADC_VP	L10	simulation	PIN54	B34_L25	U7	1.5V
PIN55	NC	-	NC	PIN56	B34_L24_P	W9	1.5V
PIN57	NC	-	NC	PIN58	B34_L24_N	Y9	1.5V
PIN59	GND	-	GND	PIN60	GND	-	GND
PIN61	B16_L1_N	F14	3.3V	PIN62	NC	-	NC
PIN63	B16_L1_P	F1 3	3.3V	PIN64	NC	-	NC
PIN65	B16_L4_N	E14	3.3V	PIN66	NC	-	NC
PIN67	B16_L4_P	E1 3	3.3V	PIN68	NC	-	NC
PIN69	GND	-	GND	PIN70	GND	-	GND
PIN71	B16_L6_N	D15	3.3V	PIN72	NC	-	NC
PIN73	B16_L6_P	D1 4	3.3V	PIN74	NC	-	NC

CON1 pin	Signal name	FPGA pin number	Level Standard	CON1 pin	Signal name	FPGA pin number	Level Standard
PIN75	B16_L8_P	C13	3.3V	PIN76	NC	-	NC
PIN77	B16_L8_N	B13	3.3V	PIN78	NC	-	NC
PIN79	NC	-	NC	PIN80	NC	-	NC

Table 2.10.1 - Expansion Port CON1 Pin Assignment

Expansion port CON 2

The 80-pin connector CON2 is used to expand the general-purpose I/O pins of FPGA BANK13 and BANK14. Both BANKs are rated for 3.3V. The pin assignments for the CON2 expansion port are shown in Table 2.10.2 .

CON2 pin	Signal name	FPGA pin number	Level Standard	CON2 pin	Signal name	FPGA pin number	Level Standard
PIN1	B13_L16_P	W15	3.3V	PIN2	B14_L16_P	V17	3.3V
PIN3	B13_L16_N	W16	3.3V	PIN4	B14_L16_N	W17	3.3V
PIN5	B13_L15_P	T14	3.3V	PIN6	B13_L14_P	U15	3.3V
PIN7	B13_L15_N	T15	3.3V	PIN8	B13_L14_N	V15	3.3V
PIN9	GND	-	GND	PIN10	GND	-	GND
PIN11	B13_L13_P	V13	3.3V	PIN12	B14_L10_P	AB21	3.3V
PIN13	B13_L13_N	V14	3.3V	PIN14	B14_L10_N	AB22	3.3V
PIN15	B13_L12_P	W11	3.3V	PIN16	B14_L8_N	AA21	3.3V
PIN17	B13_L12_N	W12	3.3V	PIN18	B14_L8_P	AA20	3.3V
PIN19	GND	-	GND	PIN20	GND	-	GND
PIN21	B13_L11_P	Y11	3.3V	PIN22	B14_L15_N	AB20	3.3V
PIN23	B13_L11_N	Y12	3.3V	PIN24	B14_L15_P	AA19	3.3V
PIN25	B13_L10_P	V10	3.3V	PIN26	B14_L17_P	AA18	3.3V
PIN27	B13_L10_N	W10	3.3V	PIN28	B14_L17_N	AB18	3.3V
PIN29	GND	-	GND	PIN30	GND	-	GND
PIN31	B13_L9_N	AA11	3.3V	PIN32	B14_L6_N	T20	3.3V
PIN33	B13_L9_P	AA10	3.3V	PIN34	B13_IO0	Y17	3.3V
PIN35	B13_L8_N	AB10	3.3V	PIN36	B14_L7_N	W22	3.3V
PIN37	B13_L8_P	AA9	3.3V	PIN38	B14_L7_P	W21	3.3V
PIN39	GND	-	GND	PIN40	GND	-	GND

CON2 pin	Signal name	FPGA pin number	Level Standard	CON2 pin	Signal name	FPGA pin number	Level Standard
PIN41	B14_L11_N	V20	3.3V	PIN42	B14_L4_P	T21	3.3V
PIN43	B14_L11_P	U20	3.3V	PIN44	B14_L4_N	U21	3.3V
PIN45	B14_L14_N	V19	3.3V	PIN46	B14_L9_P	Y21	3.3V
PIN47	B14_L14_P	V18	3.3V	PIN48	B14_L9_N	Y22	3.3V
PIN49	GND	-	GND	PIN50	GND	-	GND
PIN51	B14_L5_N	R19	3.3V	PIN52	B14_L12_N	W20	3.3V
PIN53	B14_L5_P	P19	3.3V	PIN54	B14_L12_P	W19	3.3V
PIN55	B14_L18_N	U18	3.3V	PIN56	B14_L13_N	Y19	3.3V
PIN57	B14_L18_P	U17	3.3V	PIN58	B14_L13_P	Y18	3.3V
PIN59	GND	-	GND	PIN60	GND	-	GND
PIN61	B13_L17_P	T16	3.3V	PIN62	B14_L3_N	V22	3.3V
PIN63	B13_L17_N	U16	3.3V	PIN64	B14_L3_P	U22	3.3V
PIN65	B14_L21_N	P17	3.3V	PIN66	B14_L20_N	T18	3.3V
PIN67	B14_L21_P	N17	3.3V	PIN68	B14_L20_P	R18	3.3V
PIN69	GND	-	GND	PIN70	GND	-	GND
PIN71	B14_L22_P	P15	3.3V	PIN72	B14_L19_N	R14	3.3V
PIN73	B14_L22_N	R16	3.3V	PIN74	B14_L19_P	P14	3.3V
PIN75	B14_L24_N	R17	3.3V	PIN76	B14_L23_P	N13	3.3V
PIN77	B14_L24_P	P16	3.3V	PIN78	B14_L23_N	N14	3.3V
PIN79	B14_IO0	P20	3.3V	PIN80	B14_IO25	N15	3.3V

Table 2.10.2 - Expansion Port CON2 Pin Assignment

Expansion port CON 3

The 80-pin connector CON3 is used to expand the general-purpose I/O pins of FPGA BANK15 and BANK16. Additionally, four JTAG signals are connected to the carrier board via CON3. The voltage standards of BANK15 and BANK16 can be adjusted using an LDO chip. The default installed LDO is 3.3V. If the user wants to output other standard levels, a suitable LDO can be replaced. The pin assignment of the CON3 expansion port is shown in Table 2.10.3 .

CON3 pin	Signal name	FPGA pin number	Level Standard	CON3 pin	Signal name	FPGA pin number	Level Standard
PIN1	B15_IO0	J16	3.3V	PIN2	B15_IO25	M17	3.3V
PIN3	B16_IO0	F15	3.3V	PIN4	B16_IO25	F21	3.3V

CON3 pin	Signal name	FPGA pin number	Level Standard	CON3 pin	Signal name	FPGA pin number	Level Standard
PIN5	B15_L4_P	G17	3.3V	PIN6	B16_L21_N	A21	3.3V
PIN7	B15_L4_N	G18	3.3V	PIN8	B16_L21_P	B21	3.3V
PIN9	GND	-	GND	PIN10	GND	-	GND
PIN11	B15_L2_P	G15	3.3V	PIN12	B16_L23_P	E21	3.3V
PIN13	B15_L2_N	G16	3.3V	PIN14	B16_L23_N	D21	3.3V
PIN15	B15_L12_P	J19	3.3V	PIN16	B16_L22_P	E22	3.3V
PIN17	B15_L12_N	H19	3.3V	PIN18	B16_L22_N	D22	3.3V
PIN19	GND	-	GND	PIN20	GND	-	GND
PIN21	B15_L11_P	J20	3.3V	PIN22	B16_L24_P	G21	3.3V
PIN23	B15_L11_N	J21	3.3V	PIN24	B16_L24_N	G22	3.3V
PIN25	B15_L1_N	G13	3.3V	PIN26	B15_L8_N	G20	3.3V
PIN27	B15_L1_P	H13	3.3V	PIN28	B15_L8_P	H20	3.3V
PIN29	GND	-	GND	PIN30	GND	-	GND
PIN31	B15_L5_P	J15	3.3V	PIN32	B15_L7_N	H22	3.3V
PIN33	B15_L5_N	H15	3.3V	PIN34	B15_L7_P	J22	3.3V
PIN35	B15_L3_N	H14	3.3V	PIN36	B15_L9_P	K21	3.3V
PIN37	B15_L3_P	J14	3.3V	PIN38	B15_L9_N	K22	3.3V
PIN39	GND	-	GND	PIN40	GND	-	GND
PIN41	B15_L19_P	K13	3.3V	PIN42	B15_L15_N	M22	3.3V
PIN43	B15_L19_N	K14	3.3V	PIN44	B15_L15_P	N22	3.3V
PIN45	B15_L20_P	M13	3.3V	PIN46	B15_L6_N	H18	3.3V
PIN47	B15_L20_N	L13	3.3V	PIN48	B15_L6_P	H17	3.3V
PIN49	GND	-	GND	PIN50	GND	-	GND
PIN51	B15_L14_P	L19	3.3V	PIN52	B15_L13_N	K19	3.3V
PIN53	B15_L14_N	L20	3.3V	PIN54	B15_L13_P	K18	3.3V
PIN55	B15_L21_P	K17	3.3V	PIN56	B15_L10_P	M21	3.3V
PIN57	B15_L21_N	J17	3.3V	PIN58	B15_L10_N	L21	3.3V
PIN59	GND	-	GND	PIN60	GND	-	GND
PIN61	B15_L23_P	L16	3.3V	PIN62	B15_L18_P	N20	3.3V

CON3 pin	Signal name	FPGA pin number	Level Standard	CON3 pin	Signal name	FPGA pin number	Level Standard
PIN63	B15_L23_N	K16	3.3V	PIN64	B15_L18_N	M20	3.3V
PIN65	B15_L22_P	L14	3.3V	PIN66	B15_L17_N	N19	3.3V
PIN67	B15_L22_N	L15	3.3V	PIN68	B15_L17_P	N18	3.3V
PIN69	GND	-	GND	PIN70	GND	-	GND
PIN71	B15_L24_P	M15	3.3V	PIN72	B15_L16_P	M18	3.3V
PIN73	B15_L24_N	M16	3.3V	PIN74	B15_L16_N	L18	3.3V
PIN75	NC	-		PIN76	NC	-	
PIN77	FPGA_TCK	V12	3.3V	PIN78	FPGA_TDI	R13	3.3V
PIN79	FPGA_TDO	U13	3.3V	PIN80	FPGA_TMS	T13	3.3V

Table 2.10.3 - Expansion Port CON3 Pin Assignment

Expansion port CON 4

The 80-pin connector CON4 is used to expand the general-purpose I/O of the FPGA's BANK16 and the high-speed data and clock signals of the GTP. The voltage standard of the BANK16 I/O ports can be adjusted using an LDO chip. The default installed LDO is 3.3V. If the user wants to output other standard levels, a suitable LDO can be replaced. The high-speed data and clock signals of the GTP are strictly differentially routed on the SoM, with data lines of equal length and maintaining a certain spacing to prevent signal interference. The pin assignment of the CON4 expansion port is shown in Table 2.10.4 .

CON4 pin	Signal name	FPGA pin number	Level Standard	CON4 pin	Signal name	FPGA pin number	Level Standard
PIN1	NC		NC	NC		NC	NC
PIN3	NC		NC	NC		NC	NC
PIN5	NC		NC	NC		NC	NC
PIN7	NC		NC	NC		NC	NC
PIN9	GND	-	GND	PIN10	GND	-	GND
PIN11	NC		NC	PIN12	MGT_TX2_P	B6	Differential
PIN13	NC		NC	PIN14	MGT_TX2_N	A6	Differential
PIN15	GND	-	GND	PIN16	GND	-	GND
PIN17	MGT_TX3_P	D7	Differential	PIN18	MGT_RX2_P	B10	Differential
PIN19	MGT_TX3_N	C7	Differential	PIN20	MGT_RX2_N	A10	Differential
PIN21	GND	-	GND	PIN22	GND	-	GND
PIN23	MGT_RX3_P	D9	Differential	PIN24	MGT_TX0_P	B4	Differential

CON4 pin	Signal name	FPGA pin number	Level Standard	CON4 pin	Signal name	FPGA pin number	Level Standard
PIN25	MGT_RX3_N	C9	Differential	PIN26	MGT_TX0_N	A4	Differential
PIN27	GND	-	GND	PIN28	GND	-	GND
PIN29	MGT_TX1_P	D5	Differential	PIN30	MGT_RX0_P	B8	Differential
PIN31	MGT_TX1_N	C5	Differential	PIN32	MGT_RX0_N	A8	Differential
PIN33	GND	-	GND	PIN34	GND	-	GND
PIN35	MGT_RX1_P	D11	Differential	PIN36	MGT_CLK1_P	F10	Differential
PIN37	MGT_RX1_N	C11	Differential	PIN38	MGT_CLK1_N	E10	Differential
PIN39	GND	-	GND	PIN40	GND	-	GND
PIN41	B16_L5_P	E16	3.3V	PIN42	B16_L2_P	F16	3.3V
PIN43	B16_L5_N	D16	3.3V	PIN44	B16_L2_N	E17	3.3V
PIN45	B16_L7_P	B15	3.3V	PIN46	B16_L3_P	C14	3.3V
PIN47	B16_L7_N	B16	3.3V	PIN48	B16_L3_N	C15	3.3V
PIN49	GND	-	GND	PIN50	GND	-	GND
PIN51	B16_L9_P	A15	3.3V	PIN52	B16_L10_P	A13	3.3V
PIN53	B16_L9_N	A16	3.3V	PIN54	B16_L10_N	A14	3.3V
PIN55	B16_L11_P	B17	3.3V	PIN56	B16_L12_P	D17	3.3V
PIN57	B16_L11_N	B18	3.3V	PIN58	B16_L12_N	C17	3.3V
PIN59	GND	-	GND	PIN60	GND	-	GND
PIN61	B16_L13_P	C18	3.3V	PIN62	B16_L14_P	E19	3.3V
PIN63	B16_L13_N	C19	3.3V	PIN64	B16_L14_N	D19	3.3V
PIN65	B16_L15_P	F18	3.3V	PIN66	B16_L16_P	B20	3.3V
PIN67	B16_L15_N	E18	3.3V	PIN68	B16_L16_N	A20	3.3V
PIN69	GND	-	GND	PIN70	GND	-	GND
PIN71	B16_L17_P	A18	3.3V	PIN72	B16_L18_P	F19	3.3V
PIN73	B16_L17_N	A19	3.3V	PIN74	B16_L18_N	F20	3.3V
PIN75	B16_L19_P	D20	3.3V	PIN76	B16_L20_P	C22	3.3V
PIN77	B16_L19_N	C20	3.3V	PIN78	B16_L20_N	B22	3.3V
PIN79	NC	-		PIN80	NC	-	

Table 2.10.4 - Expansion Port CON4 Pin Assignment

2.11 Power Supply

The AC7200 SoM is powered by DC 5V. When used alone, it is powered via the J3 interface; when connected to a carrier board, it is powered via the carrier board. Please ensure that J3 and the carrier board are not powered simultaneously to avoid damaging the development board. The power supply design diagram of the SoM is shown in Figure 2.11.1.

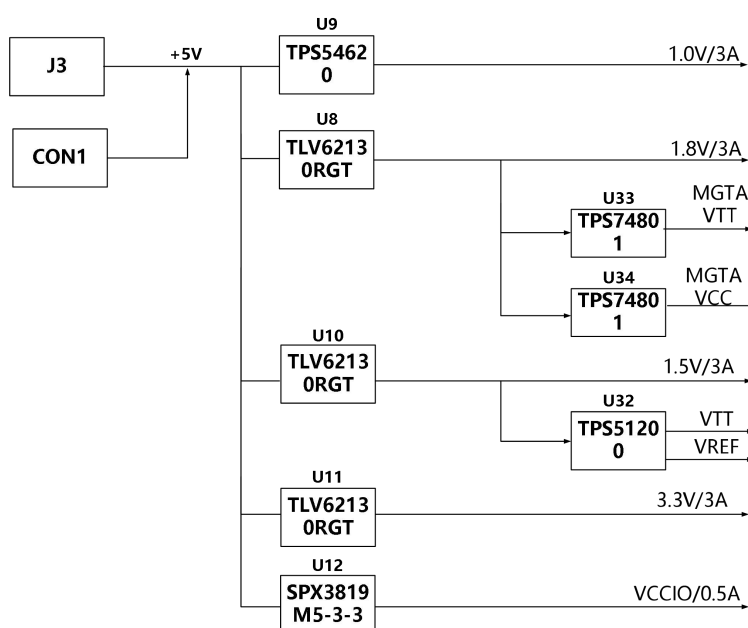


Figure 2.11.1 - Power interface section in the schematic diagram

The development board is powered by +5V, which is converted into four power supplies (+3.3V, +1.5V, +1.8V, and +1.0V) by four DC/DC power chips, each with an output current of up to 3A. A single LDO, SPX3819M5-3-3, generates the VCCIO power supply. VCCIO primarily powers the FPGA's BANK15 and BANK16. By replacing the LDO chip, the BANK15 and BANK16 I/O pins can be adapted to different voltage standards. The 1.5V power supply uses a TI TPS51200 to generate the VTT and VREF voltages required for DDR3. The 1.8V power supply uses a TI TPS74801 chip to generate the GTP transceiver power supplies MGTAVTT and MGTAVCC. The functions of each power supply are shown in the table below.

Power supply	Function
+3.3V	FPGA Bank0, Bank13, and Bank14's VCCIO, QSPI FLASH, and Clock crystal oscillator
+1.8V	FPGA auxiliary voltage, TPS74801 power supply
+1.0V	FPGA core voltage
+1.5V	DDR3, FPGA Bank34 and Bank35
VREF, VTT	DDR3
VCCIO	FPGA Bank15, Bank16
MGTAVTT	FPGA GTP transceiver Bank216

Power supply	Function
MGTAVCC	FPGA GTP transceiver Bank216

Table 2.11.1 - Power Supply and Functions

Because the Artix-7 FPGA has power-on sequence requirements, we have designed the circuit according to the chip's power requirements, with the power-on sequence being 1.0V->1.8V-> (1.5V , 3.3V , VCCIO) and 1.0V-> MGTAVCC- > MGTAVTT, to ensure the normal operation of the chip.

2.12 Structural diagram

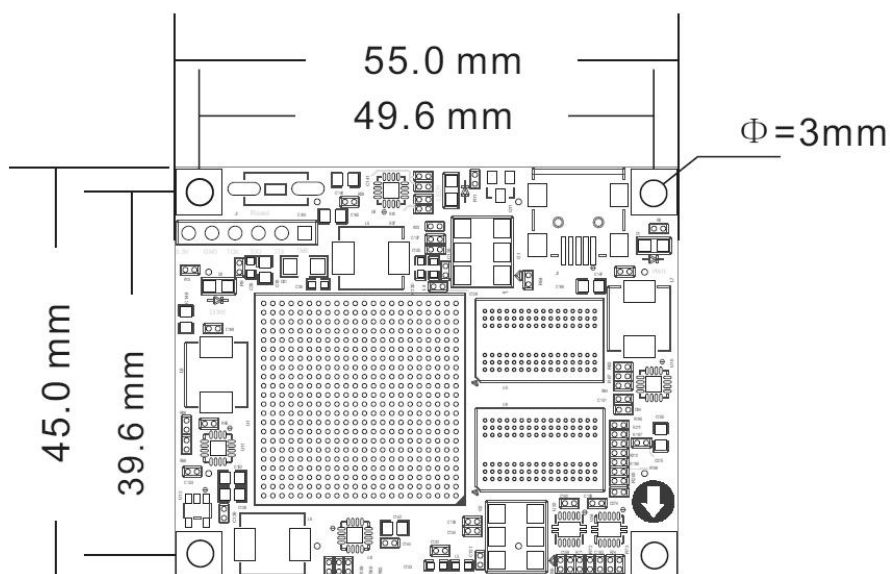


Figure 2.12.2 - Top View

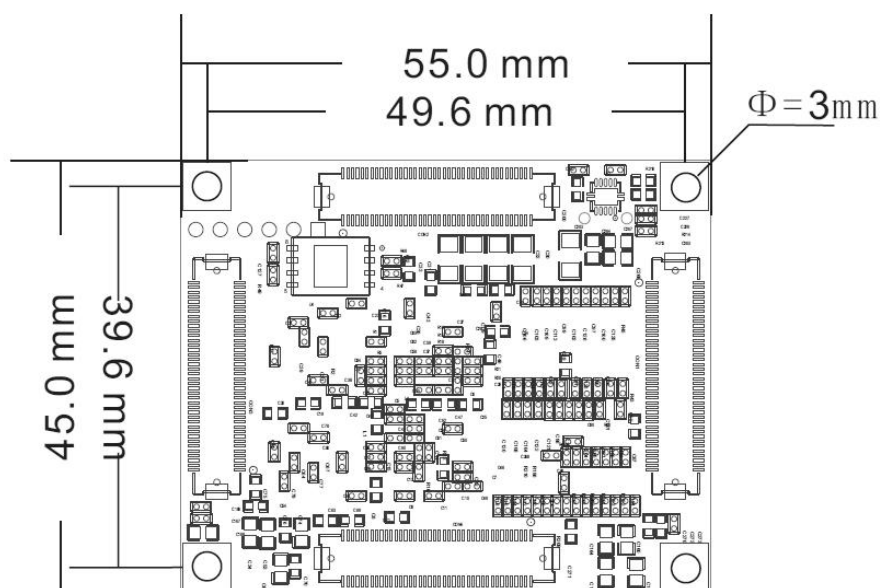


Figure 2.12.2 - Back View

Chapter 3 Carrier Board

3.1 Introduction

This chapter will provide a detailed introduction to the following interfaces and functions.

- One PCIe x4 high-speed data transmission interface
- Two 10 /100M /1000M Ethernet RJ-45 interfaces
- One HDMI video input interface
- One HDMI video output interface
- One USB UART communication interface
- SD card interface
- XADC interface
- EEPROM
- 40-pin expansion port (two-way)
- JTAG debug port
- 2 independent buttons
- 4 user LED lights

3.2 Gigabit Ethernet Interface

The AX7203 development board provides network communication services to users through two industrial-grade Ethernet GPHY chips (KSZ9031). The KSZ9031 chip supports network transmission rates of 10/100/1000 Mbps and communicates with the FPGA's MAC layer via the RGBII interface. The KSZ9031 supports MDI /MDX auto-adaptation, various speed auto-adaptation, Master/ Slave auto-adaptation, and PHY register management via the MDIO bus.

The KSZ9031 detects the level states of certain I/O pins to determine its operating mode. Table 3.2.1 describes the default settings of the GPHY chip after power-up.

Configure Pin	illustrate	Configuration value
RXD3_ADR0 RXC_ADR1 RXCTL_ADR2	PHY address in MDIO/MDC mode	PHY Address is 001
RXD1_TXDLY	TX clock 2ns delay	Delay
RXD0_RXDLY	RX clock 2ns delay	Delay

Table 3.2.1 - Default Configuration Values for PHY Chip

When the network is connected to Gigabit Ethernet, the data transmission between the FPGA and the PHY chip KSZ9031 is via the RGMII bus with a transmission clock of 125MHz . Data is sampled on the rising and falling edges of the clock .

When the network is connected to a 100 Mbps Ethernet network, data transmission between the FPGA and the KSZ9031 PHY chip is achieved via the RGMII bus with a transmission clock of 25 MHz. Data is sampled on both the rising and falling edges of the clock . Figure 3.2.1 shows a schematic diagram of the connection between the FPGA and two Ethernet PHY chips .

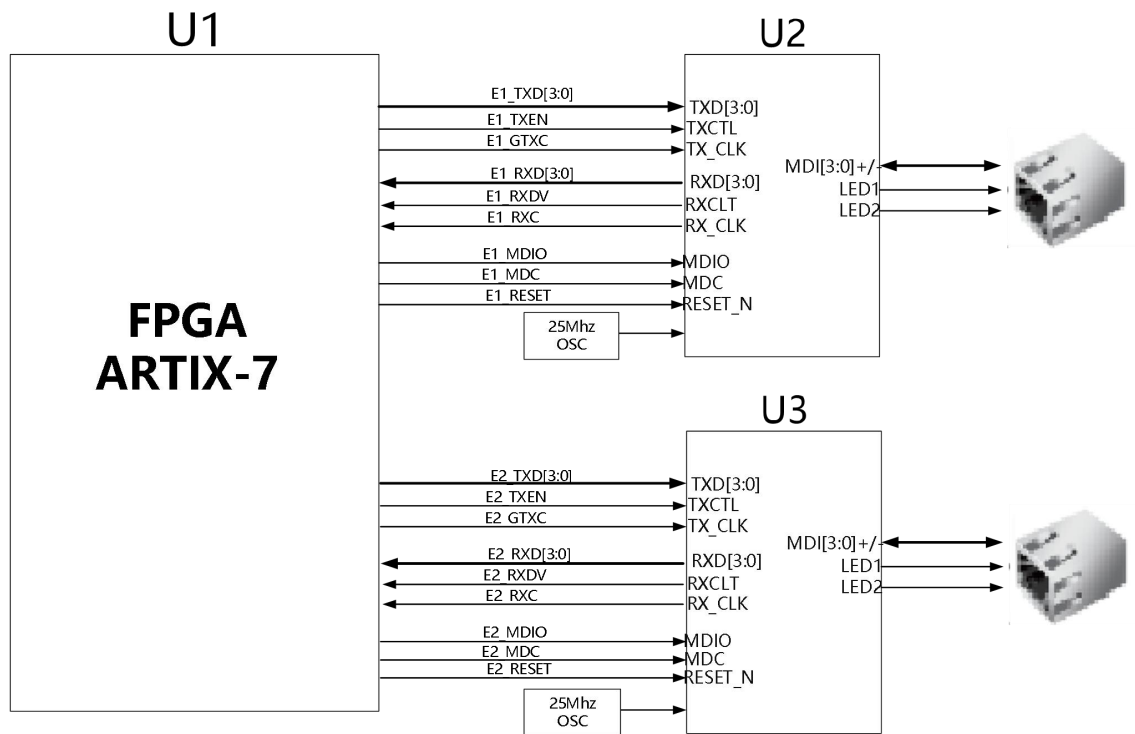


Figure 3.2.1 - Schematic diagram of FPGA and PHY chip connection

The FPGA pin assignments for Ethernet PHY1 are as follows:

Signal name	FPGA pin numbers	Remark
E1_GTXC	E18	RGMII Transmit Clock
E1_TXD0	C20	Send data bit 0
E1_TXD1	D20	Send data bit1
E1_TXD2	A19	Send data bit2
E1_TXD3	A18	Send data bit 3
E1_TXEN	F18	Send enable signal
E1_RXC	B17	RGMII Receive Clock
E1_RXD0	A16	Receive data Bit0
E1_RXD1	B18	Receive data Bit1

Signal name	FPGA pin numbers	Remark
E1_RXD2	C18	Receive data Bit2
E1_RXD3	C19	Receive data Bit3
E1_RXDV	A15	Received data valid signal
E1_MDC	B16	MDIO manages the clock
E1_MDIO	B15	MDIO Management Data
E1_RESET	D16	PHY chip reset

Table 3.2.2 - FPGA Pin Assignments for Ethernet PHY1

The FPGA pin assignments for Ethernet PHY2 are as follows:

Signal name	FPGA pin numbers	Remark
E2_GTXC	A14	RGMII Transmit Clock
E2_TXD0	E17	Send data bit 0
E2_TXD1	C14	Send data bit1
E2_TXD2	C15	Send data bit2
E2_TXD3	A13	Send data bit 3
E2_TXEN	D17	Send enable signal
E2_RXC	E19	RGMII Receive Clock
E2_RXD0	A20	Receive data Bit0
E2_RXD1	B20	Receive data Bit1
E2_RXD2	D19	Receive data Bit2
E2_RXD3	C17	Receive data Bit3
E2_RXDV	F19	Received data valid signal
E2_MDC	F20	MDIO Management the clock
E2_MDIO	C22	MDIO Management Data
E2_RESET	B22	PHY chip reset

Table 3.2.3 - FPGA Pin Assignments for Ethernet PHY2

3.3 PCIe x4 Interface

The AX7203 carrier board provides an industrial-grade high-speed data transmission PCIe x4 interface. The PCIe card's form factor conforms to the standard PCIe card electrical specifications and can be used directly in a standard PC's x4 PCIe slot.

The PCIe interface's transmit and receive signals are directly connected to the FPGA's GTP transceiver. The four-channel TX and RX signals are connected to the FPGA in a differential signal configuration, with a single-channel communication rate up to 5G bit . The PCIe reference clock is provided to the development board by the PC's PCIe slot, and the reference clock frequency is 100MHz.

The design schematic of the PCIe interface of the development board is shown in Figure 3.3.1 , where the TX transmit signal and the reference clock CLK signal are connected in AC coupling mode.

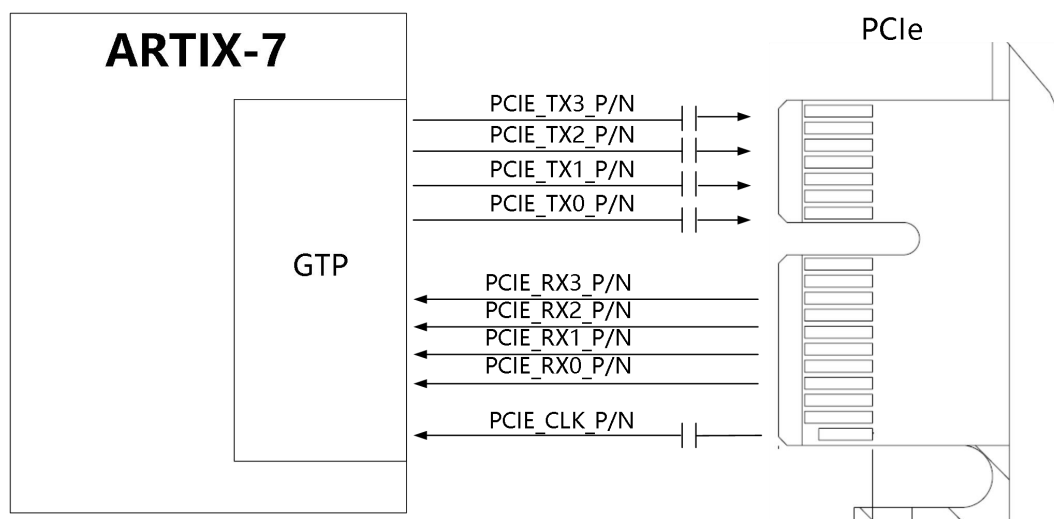


Figure 3.3.1 - Schematic diagram of PCIe x4 design

The pin assignments for the PCIe x4 interface FPGA are as follows:

Network Name	FPGA pins	Remark
PCIE _ RX0 _ P	D11	PCle Channel 0 Data Receive Positive
PCIE _ RX0 _ N	C11	PCle channel 0 data reception (Negative)
PCle _ RX1 _ P	B8	PCle Channel 1 Data Receive Positive
PCle _ RX1 _ N	A8	PCle Channel 1 Data Receiver (Negative)
PCle RX2 P	B10	PCle Channel 2 Data Receive Positive
PCle RX2 N	A10	PCle Channel 2 Data Receiver (Negative)
PCle RX3 P	D9	PCle Channel 3 Data Reception Positive
PCle _ RX3 _ N	C9	PCle Channel 3 Data Receiver (Negative)
PCIE _ TX0 _ P	D5	PCle Channel 0 Data Sending Positive
PCIE _ TX0 _ N	C5	PCle channel 0 data transmission Negative
PCle _ TX1 _ P	B4	PCle Channel 1 Data Transmission Positive
PCIE _ TX1 _ N	A4	PCle Channel 1 Data Transmission Negative

Network Name	FPGA pins	Remark
PCle _ TX2 _ P	B6	PCle Channel 2 Data Transmission Positive
PCle _ TX2 _ N	A6	PCle Channel 2 Data Transmission Negative
PCle _ TX3 _ P	D7	PCle Channel 3 Data Transmission Positive
PCle _ TX3 _ N	C7	PCle Channel 3 Data Transmission Negative
PCIE _ CLK_P	F10	PCle reference clock Positive
PCIE _ CLK_N	E10	PCle reference clock Negative
PCIE_PERST	J20	PCle reset signal

Table 3.3.1 - FPGA Pin Assignments for PCIe x4 Interface

3.4 HDMI Output Interface

The HDMI output interface is implemented using Silicon Image's SIL9134 HDMI (DVI) encoding chip, which supports up to 1080P@60Hz output and 3D output.

The SIL9134's IIC configuration interface is also connected to the FPGA's I/O. The SIL9134 is initialized and controlled through FPGA programming. The hardware connection of the HDMI output interface is shown in Figure 3.4.1 .

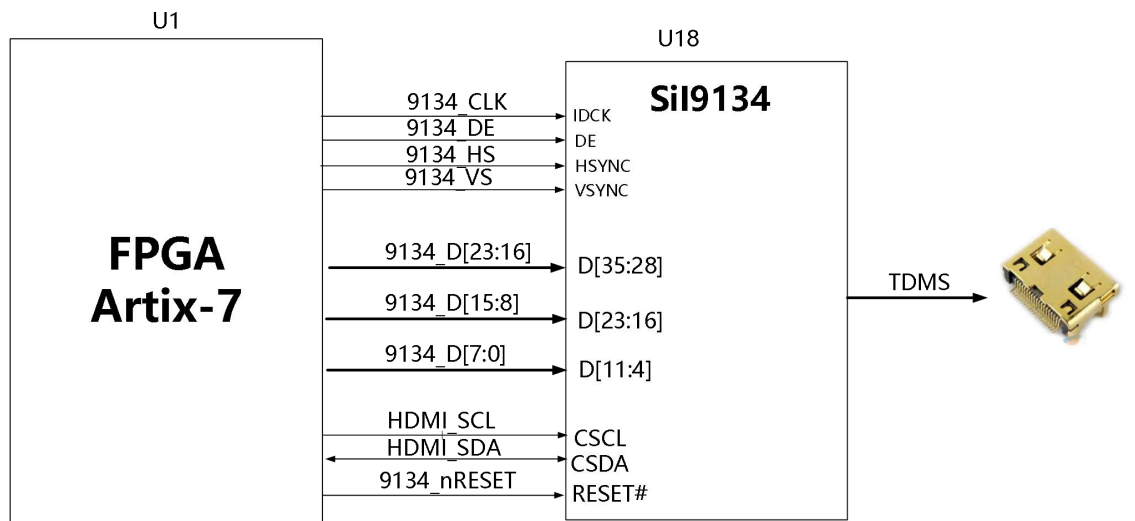


Figure 3.4.1 - Schematic diagram of HDMI output interface

FPGA pin assignment:

Pin Name	FPGA pins
9134_nRESET	J19
9134_CLK	M13
9134_HS	T15

Pin Name	FPGA pins
9134_VS	T14
9134_DE	V13
9134_D [0]	V14
9134_D [1]	H14
9134_D [2]	J14
9134_D [3]	K13
9134_D [4]	K14
9134_D [5]	L13
9134_D [6]	L19
9134_D [7]	L20
9134_D [8]	K17
9134_D [9]	J17
9134_D [10]	L16
9134_D [11]	K16
9134_D [12]	L14
9134_D [13]	L15
9134_D [14]	M15
9134_D [15]	M16
9134_D [16]	L18
9134_D [17]	M18
9134_D [18]	N18
9134_D [19]	N19
9134_D [20]	M20
9134_D [21]	N20
9134_D [22]	L21
9134_D [23]	M21

Table 3.4.1 - HDMI Output Interface Pin Assignment

3.5 HDMI Input Interface

We used Silicon Image 's SIL9013 HDMI decoding chip, which supports up to 1080P@60Hz

input and different data output formats .

The SIL9013's IIC configuration interface is also connected to the FPGA's I/O. The SIL9013 is initialized and controlled through FPGA programming. The hardware connection of the HDMI input interface is shown in Figure 3.5.1 .

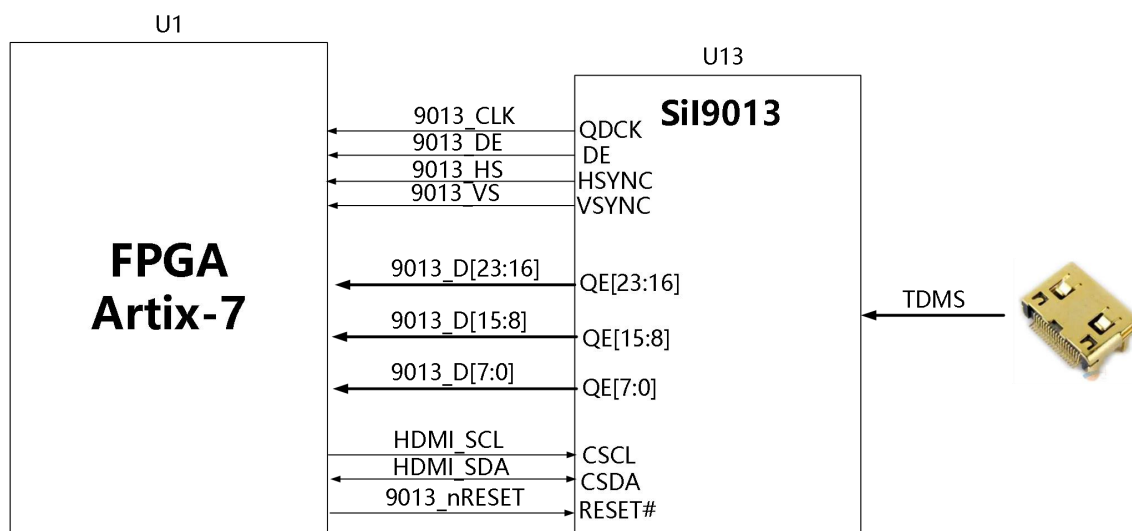


Figure 3.5.1 - HDMI Input Schematic Diagram

FPGA pin assignment:

Pin Name	FPGA pins
9013_nRESET	H19
9013_CLK	K21
9013_HS	K19
9013_VS	K18
9013_DE	H17
9013_D [0]	H18
9013_D [1]	N22
9013_D [2]	M22
9013_D [3]	K22
9013_D [4]	J22
9013_D [5]	H22
9013_D [6]	H20
9013_D [7]	G20
9013_D [8]	G22

Pin Name	FPGA pins
9013_D [9]	G21
9013_D [10]	D22
9013_D [11]	E22
9013_D [12]	D21
9013_D [13]	E21
9013_D [14]	B21
9013_D [15]	A21
9013_D [16]	F21
9013_D [17]	M17
9013_D [18]	J16
9013_D [19]	F15
9013_D [20]	G17
9013_D [21]	G18
9013_D [22]	G15
9013_D [23]	G16

Table 3.5.1 - HDMI Input Pin Assignment

3.6 SD Card slot

An SD card (Secure Digital Memory Card) is a type of memory card based on semiconductor flash memory technology. The concept was spearheaded by Panasonic of Japan in 1999, with Toshiba and SanDisk of the United States conducting the actual research and development. In 2000, these companies founded the Secure Digital Association (SDA) , a powerful consortium that attracted numerous manufacturers, including IBM , Microsoft , Motorola , NEC, and Samsung. Driven by these leading manufacturers, the SD card has become the most widely used memory card in consumer digital devices.

SD cards are commonly used storage devices. Our expanded SD card supports both SPI and SD modes, and uses a MicroSD card. The schematic diagram is shown in Figure 3.6.1 .

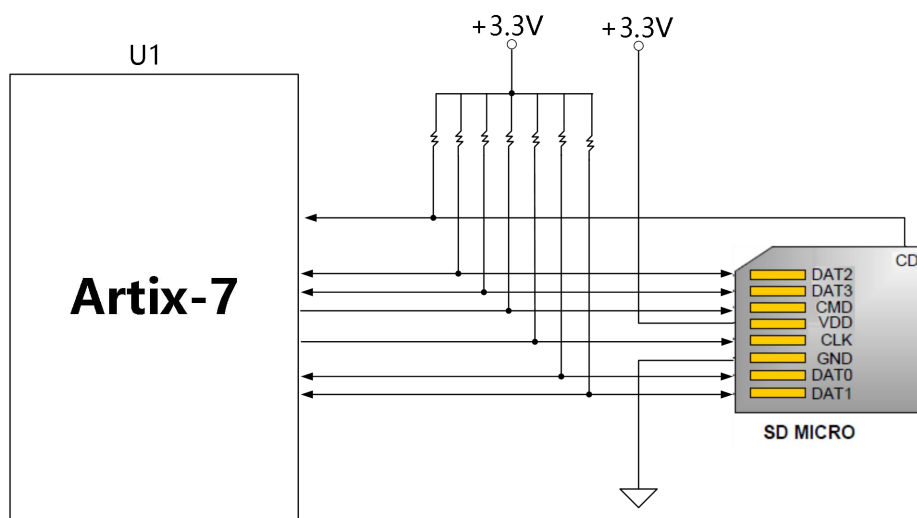


Figure 3.6.1 - SD Card Slot Schematic Diagram

SD card slot pin assignment

Pin Name	FPGA pins
SD_CLK	AB12
SD_CMD	AB11
SD_CD_N	F14
SD_DAT0	AA13
SD_DAT1	AB13
SD_DAT2	Y13
SD_DAT3	AA14

Table 3.6.1 - SD Card Pin Assignment

3.7 USB to Serial Port

The AX7203 development board includes the Silicon Labs CP2102GM USB- UART chip . The USB interface uses a MINI USB connector, allowing it to be connected to a PC 's USB port for serial data communication using a single USB cable. A schematic diagram of the USB UART circuit design is shown below :

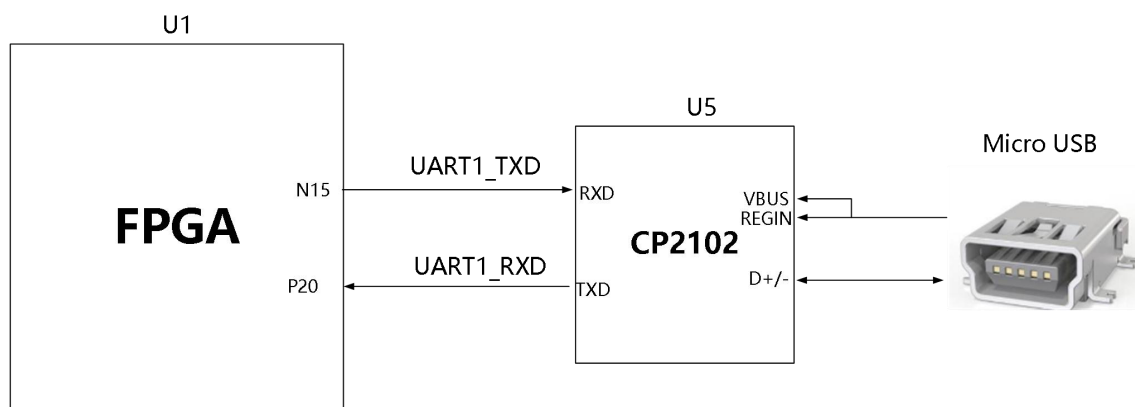


Figure 3.7.1 - Schematic diagram of USB to serial port adapter

At the same time, two LED indicators (LED3 and LED4) silkscreened as TX and RX on the PCB are set for the serial port signal. The TX and RX LEDs will indicate whether the serial port is sending or receiving data , as shown in the figure below .

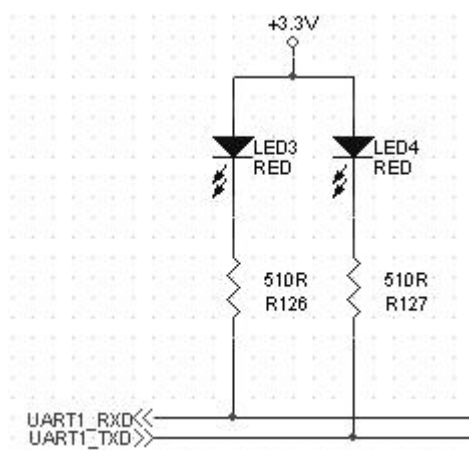


Figure 3.7.2 - USB to serial port signal indicator

UART to serial port conversion:

Pin Name	FPGA pins
UART1_RXD	P20
UART1_TXD	N15

Table 3.7.1 - UART Pin Assignment

3.8 EEPROM

The AX7203 development board features an EEPROM, model 24LC04 , with a capacity of 4Kbit (2*256*8bit), consisting of two 256-byte blocks , communicating via the I2C bus. The EEPROM is used to learn the I2C bus communication method. The EEPROM's I2C signals are connected to the BANK14 I/O port on the FPGA side. Figure 3.8.1 shows the EEPROM design schematic.

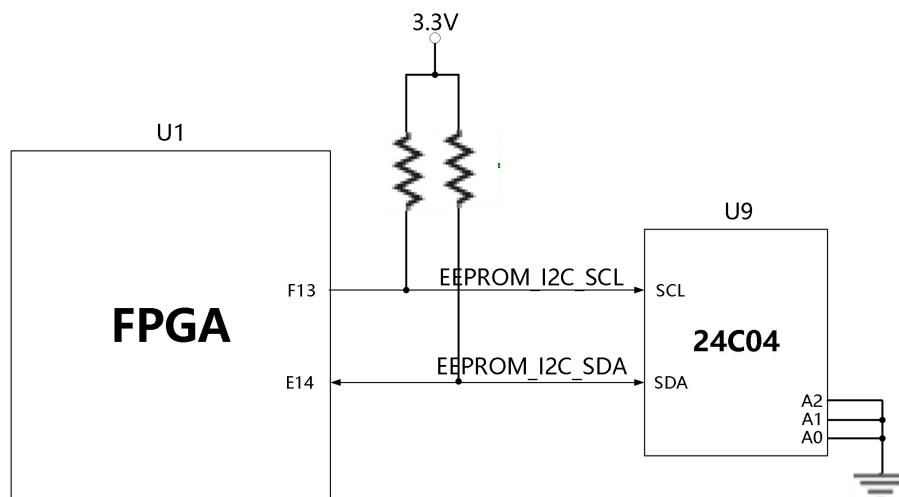


Figure 3.8.1 - EEPROM Schematic Diagram

EEPROM pin assignment:

Pin Name	FPGA pins
EEPROM_I2C_SCL	F13
EEPROM_I2C_SDA	E14

Table 3.8.1 - EEPROM Pin Assignment

3.9 Expansion Port

The carrier board has two 40-pin expansion ports, J11 and J13, with a standard 2.54mm pitch, for connecting various Alinx modules or user-designed external circuits. These expansion ports have 40 signals: one 5V power supply, two 3.3V power supply, three ground lines, and 34 I/O ports. **Do not connect the I/O ports directly to 5V devices, as this may damage the FPGA. If you need to connect 5V devices, you will need to use a level converter chip.**

A 33-ohm resistor array is connected in series between the expansion port and the FPGA to protect the FPGA from damage caused by excessive external voltage or current. The pin assignments for the J11 expansion port FPGA are shown in the table below:

Pin number	FPGA pins	Pin number	FPGA pins
1	GND	2	+5V
3	P16	4	R17
5	R16	6	P15
7	N17	8	P17
9	U16	10	T16
11	U17	12	U18

Pin number	FPGA pins	Pin number	FPGA pins
13	P19	14	R19
15	V18	16	V19
17	U20	18	V20
19	AA9	20	AB10
21	AA10	22	AA11
23	W10	24	V10
25	Y12	26	Y11
27	W12	28	W11
29	AA15	30	AB15
31	Y16	32	AA16
33	AB16	34	AB17
35	W14	36	Y14
37	GND	38	GND
39	+3.3V	40	+3.3V

Table 3.9.1 - J11 Expansion Port Pin Assignment

FPGA pin assignment for expansion port (J13) :

Pin number	FPGA pins	Pin number	FPGA pins
1	GND	2	+5V
3	W16	4	W15
5	V17	6	W17
7	U15	8	V15
9	AB21	10	AB22
11	AA21	12	AA20
13	AB20	14	AA19
15	AA18	16	AB18
17	T20	18	Y17
19	W22	20	W21
21	T21	22	U21
23	Y21	24	Y22

Pin number	FPGA pins	Pin number	FPGA pins
25	W20	26	W19
27	Y19	28	Y18
29	V22	30	U22
31	T18	32	R18
33	R14	34	P14
35	N13	36	N14
37	GND	38	GND
39	+3.3V	40	+3.3V

Table 3.9.2 - Pin Assignment for Expansion Port (J13)

3.10 JTAG Interface

The development board has a JTAG interface for downloading FPGA programs or burning programs to FLASH. To prevent damage to the FPGA chip caused by hot-plugging, we added a protection diode to the JTAG signal to ensure that the signal voltage is within the acceptable range for the FPGA, thus preventing FPGA damage. Avoid hot-plugging the JTAG cable.

3.11 XADC Interface (not installed by default)

The AX7203 carrier board expands the XADC connector interface using a 2x8 2.54mm pitch dual-row pin header. The XADC interface expands with three pairs of differential ADC input interfaces, connecting to the FPGA's 12-bit 1Msps analog-to-digital converter. One pair of differential interfaces connects to the FPGA's dedicated differential analog input channels VP/VN, and the other two pairs connect to auxiliary analog input channels (Analog Channel 0 and Analog Channel 9). The pin assignments for the XADC interface are as follows.

XADC interface	FPGA pins	Input amplitude	Describe
1, 2	VP_0 : L10 VN_0 : M9	Peak-to-peak value 1V	FPGA dedicated XADC input channel
5, 6	AD9P : J15 AD9N : H15	Peak-to-peak value 1V	FPGA -assisted XADC input channel 9 (can be used as a general IO).
9, 10	AD0P : H13 AD0N : G13	Peak-to-peak value 1V	FPGA -assisted XADC input channel 0 (can be used as a general-purpose I/O)

Table 3.11.1 Pin Assignments for the XADC Interface

3.12 Button

The carrier board contains two user buttons, KEY1 and KEY2. Both buttons are connected to the FPGA's general-purpose I/O pins. The buttons are active low; when a button is pressed, the

FPGA's I/O input voltage is low, and when no button is pressed, the FPGA's I/O input voltage is high. The button circuit is shown in section 3.12.1.

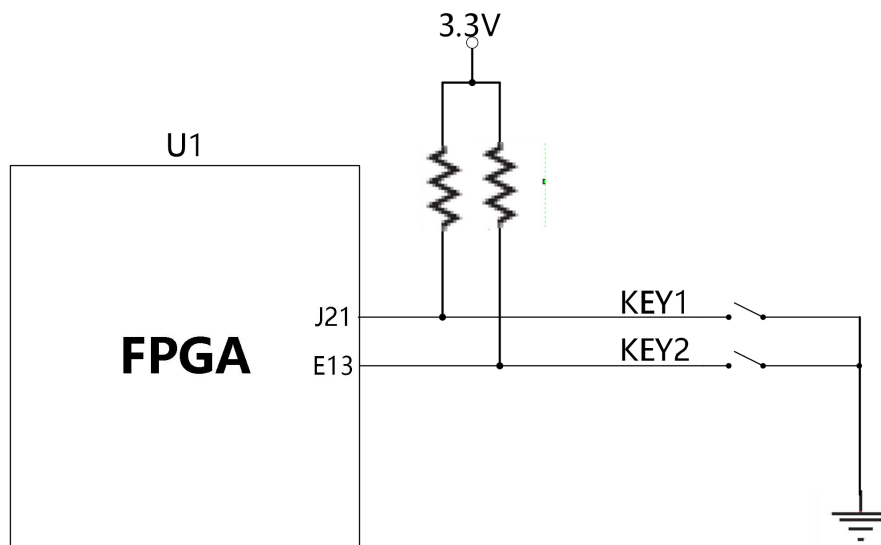


Figure 3.12.1 - Schematic diagram of button hardware design

Button FPGA Pin Assignment:

Pin Name	FPGA pins
KEY1	J21
KEY2	E13

Table 3.12.1 - FPGA Pin Assignment for Buttons

3.13 LED

The carrier board has seven red LEDs: one power indicator (PWR) , two USB UART data receive and transmit indicators, and four user LEDs (LED1-LED4). The power indicator lights up when the development board is powered on. User LEDs 1-LED4 are connected to the FPGA's general-purpose I/O pins. When the I/O voltage connected to the user LEDs is configured to a low level, the user LEDs light up; when the I/O voltage is configured to a high level, the user LEDs turn off. A schematic diagram of the LED hardware connection is shown in Figure 3.13.1.

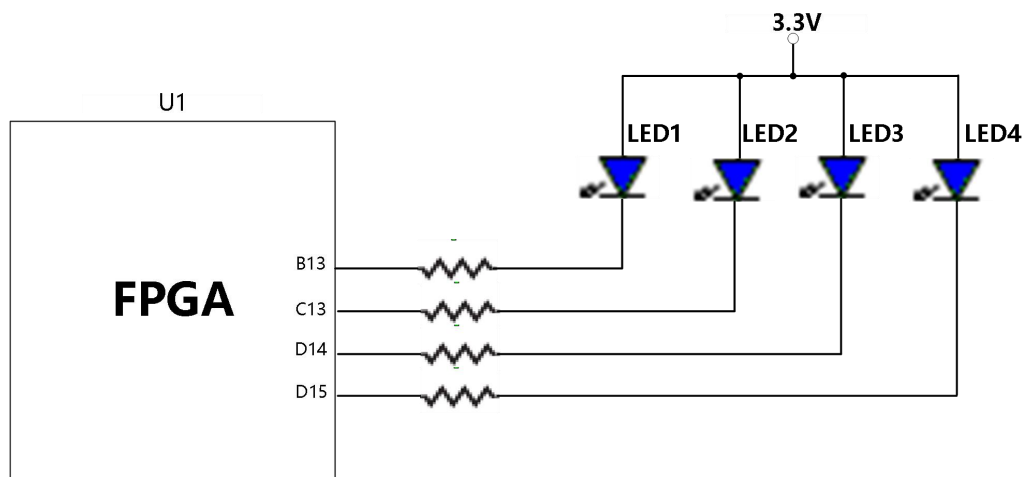


Figure 3.13.1 - Schematic diagram of LED lamp hardware design

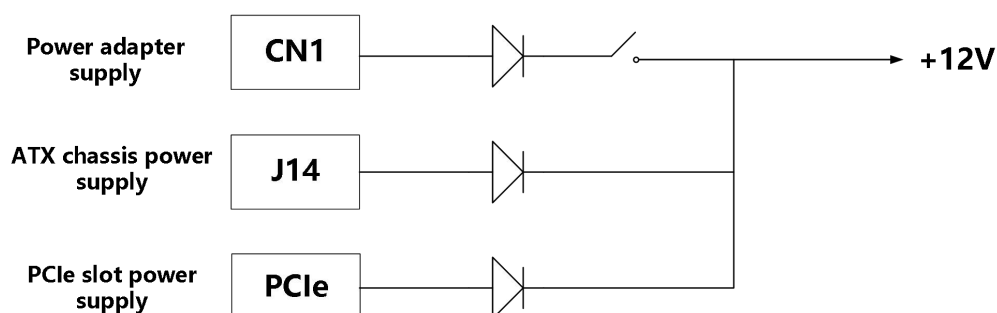
LED Lamp FPGA Pin Assignment:

Pin Name	FPGA pins
LED1	B13
LED2	C13
LED3	D14
LED4	D15

Table 3.13.1 - LED Lamp FPGA Pin Assignment

3.14 Power supply

The development board's power input voltage is DC 12V. Please use the power supply that comes with the development board and do not use other power supplies to avoid damaging the development board. The development board also supports drawing power from the PCIe interface and directly from the ATX chassis power supply (12V).



The carrier board uses four ETA1471 DC/DC power chips to convert the +12V voltage into three power supplies: +5V, +3.3V, and +1.8V. Additionally, the +5V power supply on the carrier board powers the SoM via a board-to-board connector. The power supply design on the carrier board is shown in Figure 3.14.1.

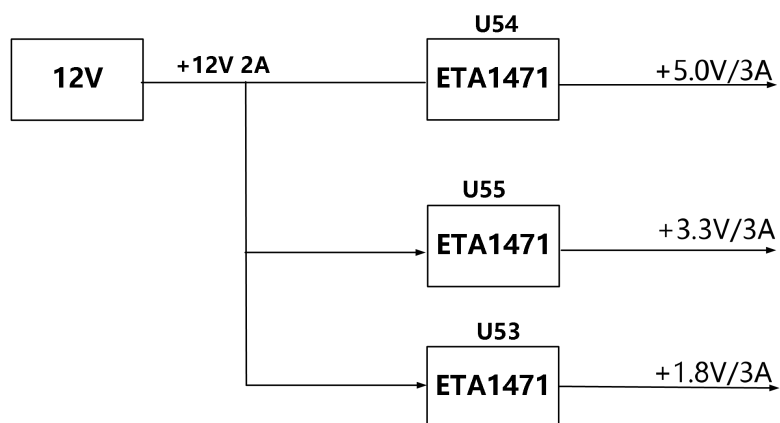


Figure 3.14.1 - Carrier board power supply schematic



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